Agentic *reasoning*.

From interacting with the world
to improving how we work together.

01 Acting → 02 Learning → 03 Collaborating

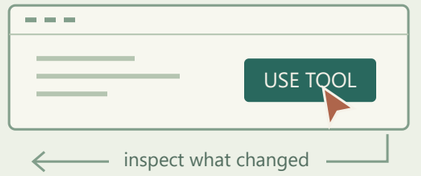
Section 6.5 Autonomous Web Exploration & Research Agents

A RESEARCH WORKBENCH § 6.5

01 / FOUNDATIONAL

Act on the task.

Tools + observations



02 / SELF-EVOLVING

Learn the lesson.

Experience → workflow



03 / COLLECTIVE

Build a shared result.

Specialists + review



Complementary capabilities, combined around one goal.

Three capabilities. One growing system.

01

Acting

Turn goals into actions. Check what changed.

FOUNDATIONAL REASONING

02

Learning

Turn experience into reusable guidance.

SELF-EVOLVING REASONING

03

Collaborating

Turn specialized work into a shared result.

COLLECTIVE REASONING

Wei et al. · Survey §6.5, pp. 57–64 · §6.5.1–6.5.3 · Capabilities can be combined

From responses to *real-world tasks.*

Find a nonstop Raleigh-to-Boston flight under $300 on your chosen date.

TRADITIONAL LLM

Suggest a flight.

Respond using the information it is given.

→

AGENTIC FLIGHT SEARCH

01 Search route & date → 02 Filter nonstop → 03 Check fare details → 04 Return options

THREE FOUNDATIONAL CAPABILITIES MAKE THIS POSSIBLE

01 **Planning**

Choose the search steps and travel constraints.

02 **Tool use**

Enter details and apply filters in a browser.

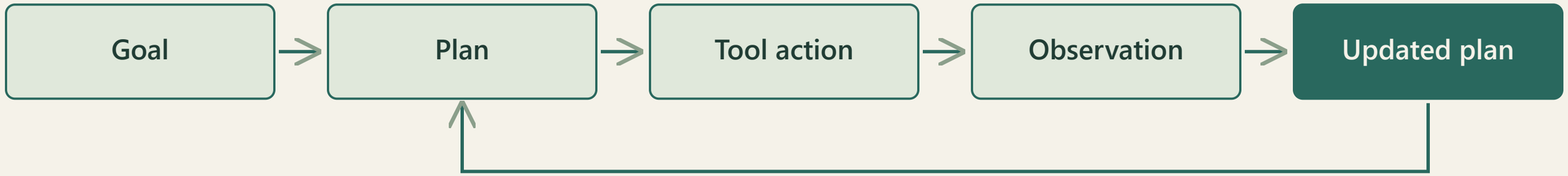
03 **Search & retrieval**

Find flights and retrieve current fare details.

Together, these capabilities connect *reasoning to execution.*

Reasoning becomes a *feedback loop*.

Find a nonstop flight from Raleigh to Boston under $300.



Environmental feedback changes the next decision.

05

Refresh the results and check another nonstop option.

Keep the same date, route, and budget.

WHERE AGENTS ACT

Same loop. Different environments.

Planning, tool use, and retrieval connect all three.



Web agents

Find and cross-check dynamic evidence.

Search, navigate, extract. Page state changes.

WebVoyager · WebThinker



GUI agents

Operate software through its interface.

Perceive, locate, click, type. Ground intent in pixels.

Agent S · OS-Copilot



Research agents

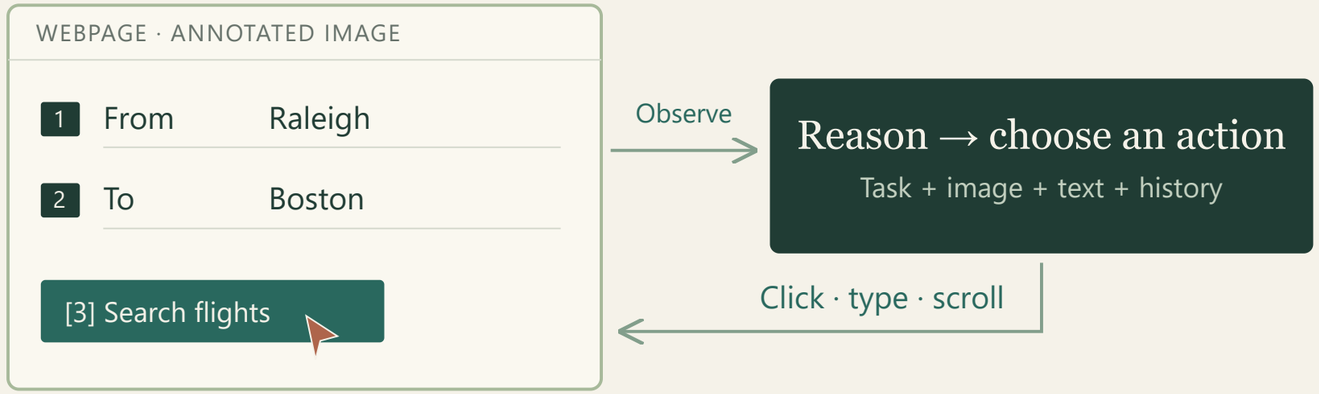
Orchestrate an inquiry from idea to report.

Retrieve, experiment, interpret. Preserve evidence and code.

Agent Laboratory · Dolphin

WebVoyager: knowing is not *executing*.

Can a multimodal model complete tasks on real websites?



Updated screenshot → next decision · stop or reach the step limit

Planning
Choose the next step.

Tool use
Operate the browser.

Search & retrieval
Locate task-relevant content.

REPORTED TASK SUCCESS

59.1%

643 tasks · 15 real websites

GPT-4V backbone · human evaluation
Whole-task success, not step accuracy.

EXECUTION REMAINS FRAGILE

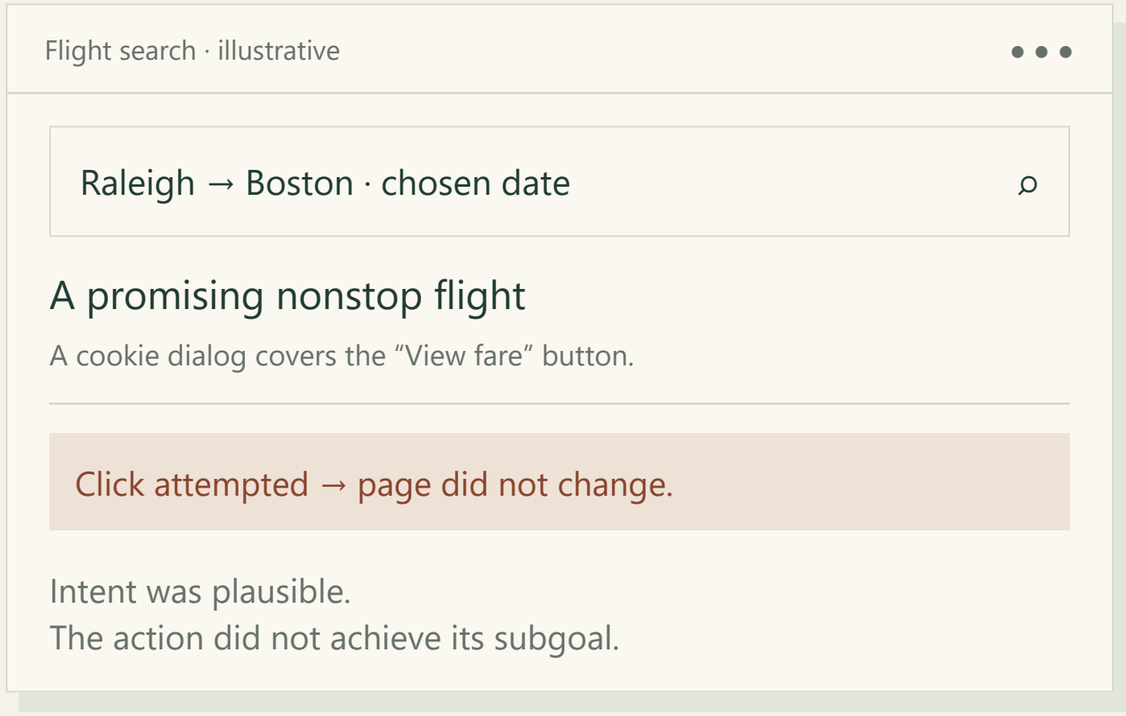
Wrong targets, misread pages, and getting stuck can derail a task.

> **Understanding the next step ≠ *carrying it out reliably*.**
>
> A model can choose a sensible action and still fail to execute it correctly.

He et al. · WebVoyager (ACL 2024) · §3; §5, Table 1 and error analysis · Survey §6.5.1, ref. [638] · Diagram adapted for teaching

THE RELIABILITY PROBLEM

A plan must survive contact with the page.



1 **Detect the mismatch**

Compare the expected state with the observed page.

2 **Repair the local action**

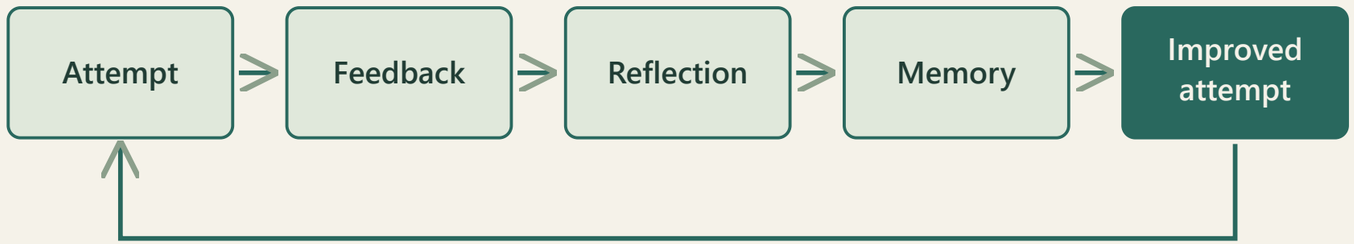
Close the dialog; locate “View fare”; retry.

3 **Preserve the lesson**

Retain a useful recovery strategy for later tasks.

Long action chains amplify small errors. *Verification makes recovery possible.*

Experience should change the *next attempt*.



Store what worked. Reuse it when the task changes.

Recall the procedure; adapt it to the new goal.

05

A FUTURE FLIGHT SEARCH

A reusable starting point.

```
search_flights(route,date)
→ filter nonstop
→ check fare & availability
```

Reuse the search steps for a new route and date. Check fares and availability again.

Remember a procedure, *not a whole episode.*

Problem: long tasks repeat useful subroutines, yet agents often start afresh.

01 / EXPERIENCE

Collect trajectories

Observations and actions from previous tasks.

→

02 / INDUCTION

Abstract a routine

Extract a reusable subtask. Replace specifics with variables.

→

03 / USE

Guide new actions

Add workflows to the prompt. Keep observing the environment.

"Find a nonstop flight from **Raleigh to Boston**."

→

```
search_flights({route}, {date})
enter → filter → verify
```

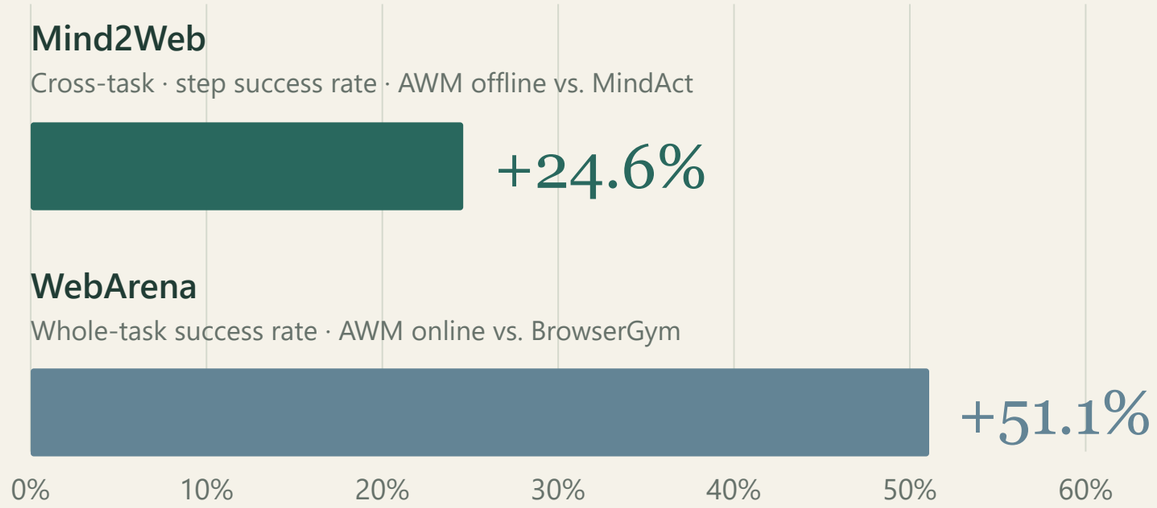
Offline: induce from training examples.

Online: update after evaluator-approved attempts.

Wang et al. · Agent Workflow Memory (2024) · §2.2–2.3; survey §6.5.2 · Example adapted for teaching

Memory helps. Read the *denominator*.

RELATIVE IMPROVEMENT OVER THE LISTED BASELINE (%)



Hover, focus, or tap a bar for the original rates.

Relative ≠ percentage points

$$\frac{(\text{AWM} - \text{baseline})}{\text{baseline}} \times 100$$

36.2% → 45.1%
+8.9 percentage points

23.5% → 35.5%
+12.0 percentage points

Different metrics and baselines.
Do not rank the benchmarks by bar height.

Wang et al. · Agent Workflow Memory (2024) · §3.1, Table 1; §3.2, Table 3 · GPT-4 · Reported rounded values

Better targeting can hide *weaker actions.*

Element selection accuracy (%)

41.6 → 50.6 MindAct → AWM

Action F_1 (reported score)

60.6 → 57.3 MindAct → AWM

Step success rises overall: 36.2% → 45.1%.
The two component metrics move in different directions.

Reusable guidance helps locate the right thing.

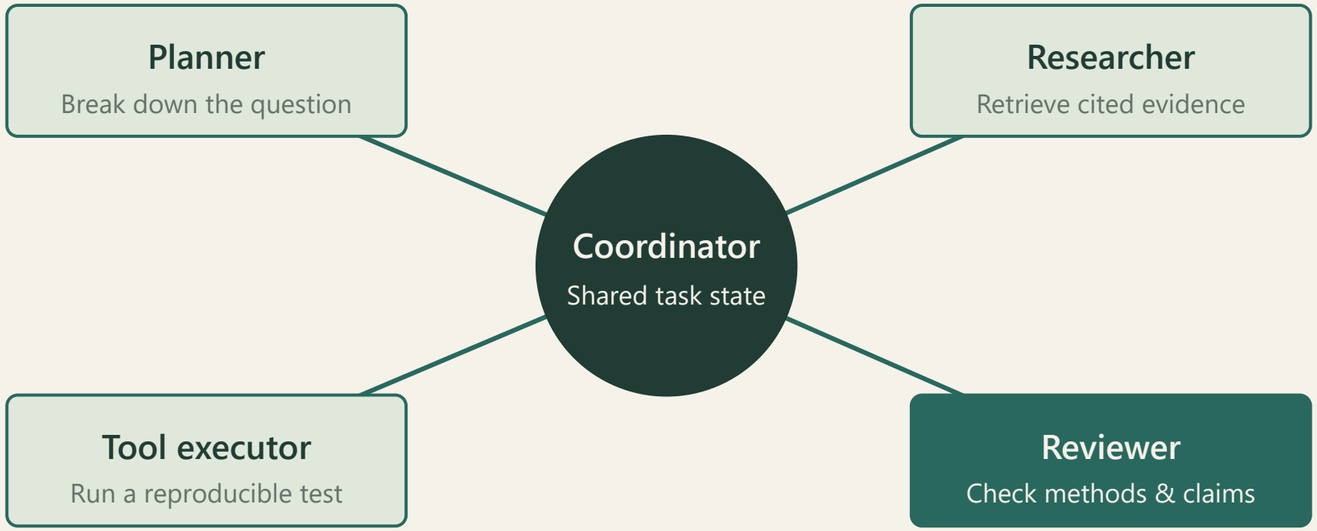
But a familiar routine may suggest an action that does not fit the current page.

> Design lesson: remember a strategy, then verify whether it still applies.

Other limits: evaluator errors, stale procedures, and low whole-task success.

Wang et al. · Agent Workflow Memory (2024) · Table 3 (GPT-4, Mind2Web cross-task); §3.2.1, §3.2.2 · Model-based evaluation can admit bad memories

Give each agent a job— and an *interface*.



Explicit roles + shared artifacts + a review gate

Challenge, then integrate.

Reviewer: challenge unsupported claims before the coordinator integrates the result.

> Useful specialization requires visible artifacts and a way to challenge them.

Survey examples: INFOGENT separates navigation, extraction, and aggregation. Mobile-Agent-E separates management, perception, action, reflection, and notes.

A collaboration is a sequence of *handoffs*.

ILLUSTRATIVE SCIENTIFIC QUESTION

Does image noise affect two model families differently?

Keep the data, perturbation, and evaluation protocol fixed.

PLANNER →	Protocol: same test set, several noise levels, repeated seeds.
RESEARCHER →	Evidence: related methods, dataset assumptions, source links.
EXECUTOR →	Artifacts: runnable code, exact settings, raw results.
REVIEWER →	Decision: check leakage, missing controls, and unsupported claims.

The coordinator integrates only what the evidence supports.
A disagreement should trigger a check, not merely a vote.

Wei et al. · Survey §6.5, pp. 57–64 · Teaching scenario inspired by §6.5.3; this is not a reported Agent Laboratory experiment

A research assistant, organized as a *lab*.

Problem: connect literature, experiments, and reporting around a human research idea.

01 / READ

Literature review

Search arXiv.
Build a grounded starting point.

PhD student agent

→

02 / TEST

Experimentation

Plan → prepare data → run code.
Reflect on errors and results.

PhD · Postdoc · ML engineer
mle-solver: propose, execute, score, revise

→

03 / EXPLAIN

Report writing

Interpret findings.
Draft and refine a report.

PhD · Professor
paper-solver: iterative report refinement

HUMAN GUIDANCE

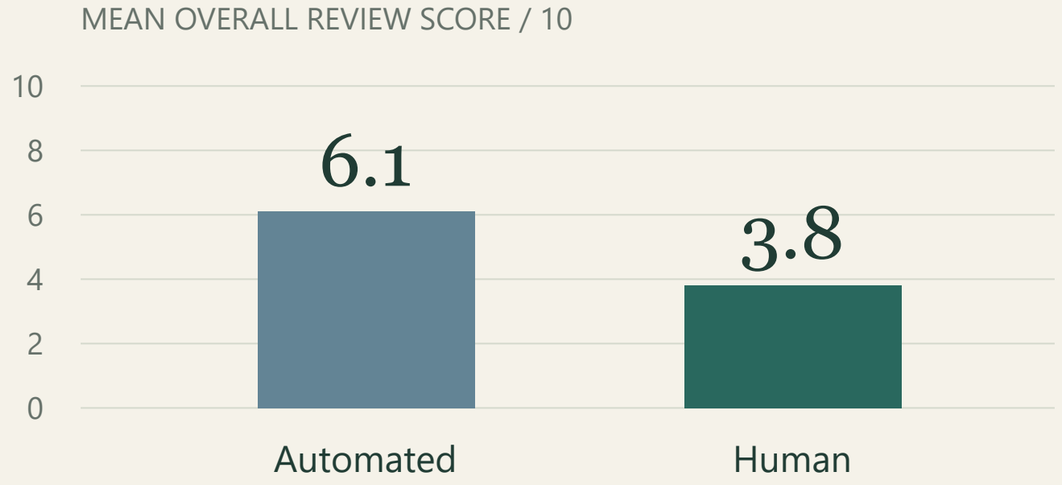
Research idea at entry · feedback at subtask boundaries in co-pilot mode

Output / research report + code repository

Persistent artifacts connect the phases.

Schmidgall et al. · Agent Laboratory (2025) · §3; survey §6.5.1–6.5.2 · Simplified architecture; actual workflow has additional substages

A convincing report can fool its *reviewer*.



Autonomous study: 5 research questions × 3 model backends = 15 papers. 10 PhD volunteers each reviewed 3 randomly assigned papers.

Review quality is a bottleneck.

Automated scores exceeded human scores by **2.3 points** on average.

> Co-pilot reports received a higher external overall score: **4.38/10** vs. **3.8/10** for autonomous reports.

Improvements were stronger in quality and presentation than in significance or contribution. Small, task-specific study; comparisons do not isolate a causal effect.

Useful assistance is plausible. *Scientific validity still needs independent checks.*

Schmidgall et al. · Agent Laboratory (2025) · §4.1–4.2; Fig. 6–7 · Scores are study averages, not acceptance probabilities

Add capabilities. Keep the *evidence loop*.

System configuration	Act with tools	Reuse experience	Coordinate specialists
Foundational	✓ Interact & verify	Optional extension	Optional extension
+ Self-evolving	✓	+ Persistent adaptation	Optional extension
+ Collective	✓	✓ Combined here	+ Shared, specialized work

Acting
Verify the observed outcome.

Learning
Check the quality of what persists.

Collaborating
Audit the artifacts and the reviewer.

Wei et al. · Survey §6.5, pp. 57–64 · Synthesis of §6.5 with AWM and Agent Laboratory · “Combined” indicates a possible design, not a universal requirement